



Discovering Emergent Soil Carbon Mechanisms with a Biogeochemistry-Informed Neural Network (BINN)

Haodi Xu¹, Feng Tao², Joshua Fan³, Carla P. Gomes³, Yiqi Luo¹
Cornell University, Department of ¹Soil & Crop Science, ²Ecology & Evolutionary Biology, ³Computer Science

Contact: Haodi Xu (hx293@cornell.edu), Joshua Fan (jyf6@cornell.edu)

Background

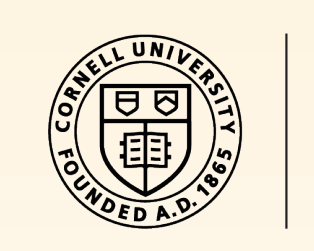
- Scientists have developed process-based models based on current scientific knowledge for more robust predictions
- Although we have large amounts of environmental data, it is difficult to use it to refine our current understanding of scientific processes
- On the other hand, machine learning can learn complex relationships from data, but does not help us understand underlying physical processes
- Our approach: combine **scientific knowledge (process-based models)** and **data-driven learning (neural networks)** into an integrated system that can be trained end-to-end

Interdisciplinary Collaborations

- This project involved close collaboration between ecologists and computer scientists
 - We needed insights from both **soil biogeochemistry** and **AI**
- This research was supported by the AI-CLIMATE Institute, Cornell Schmidt AI in Science Postdoctoral Fellowship, and NSF Digital Plant Science fellowship. Computational resources are provided by the US National Center for Atmospheric Research (NCAR). We appreciate their support for cross-cutting interdisciplinary research!



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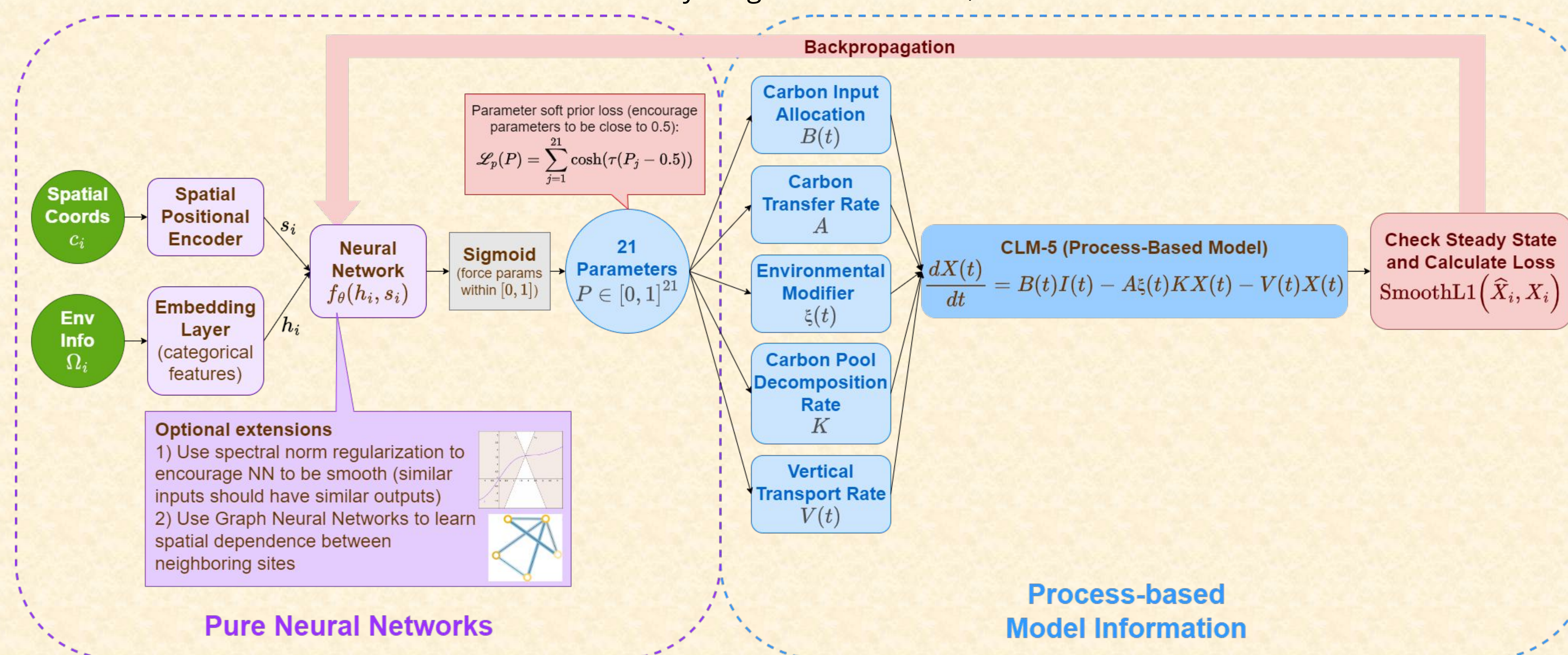
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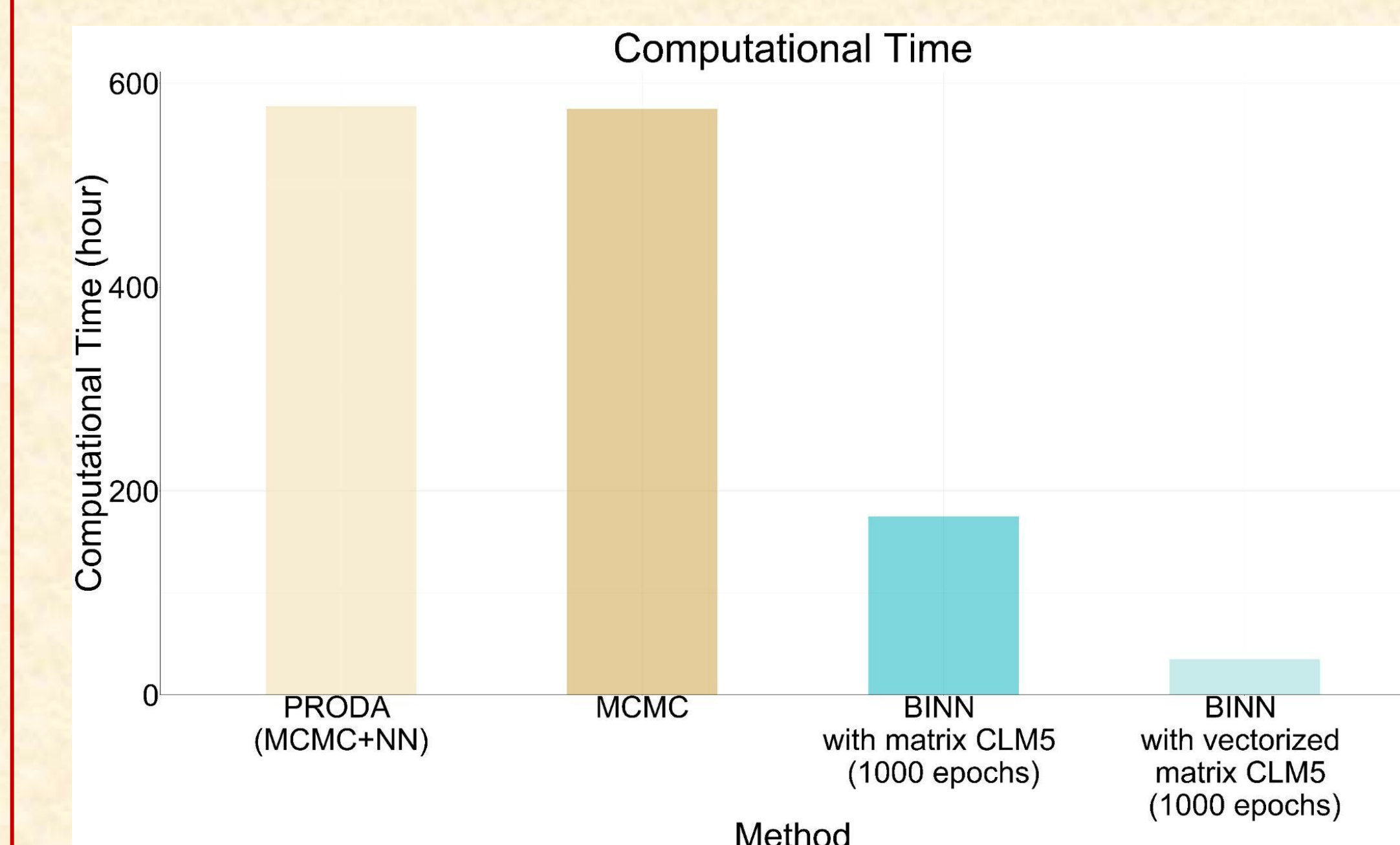
Our BINN Approach

- Embed process-based model as final layer of neural network
- Since everything is differentiable, can train end-to-end



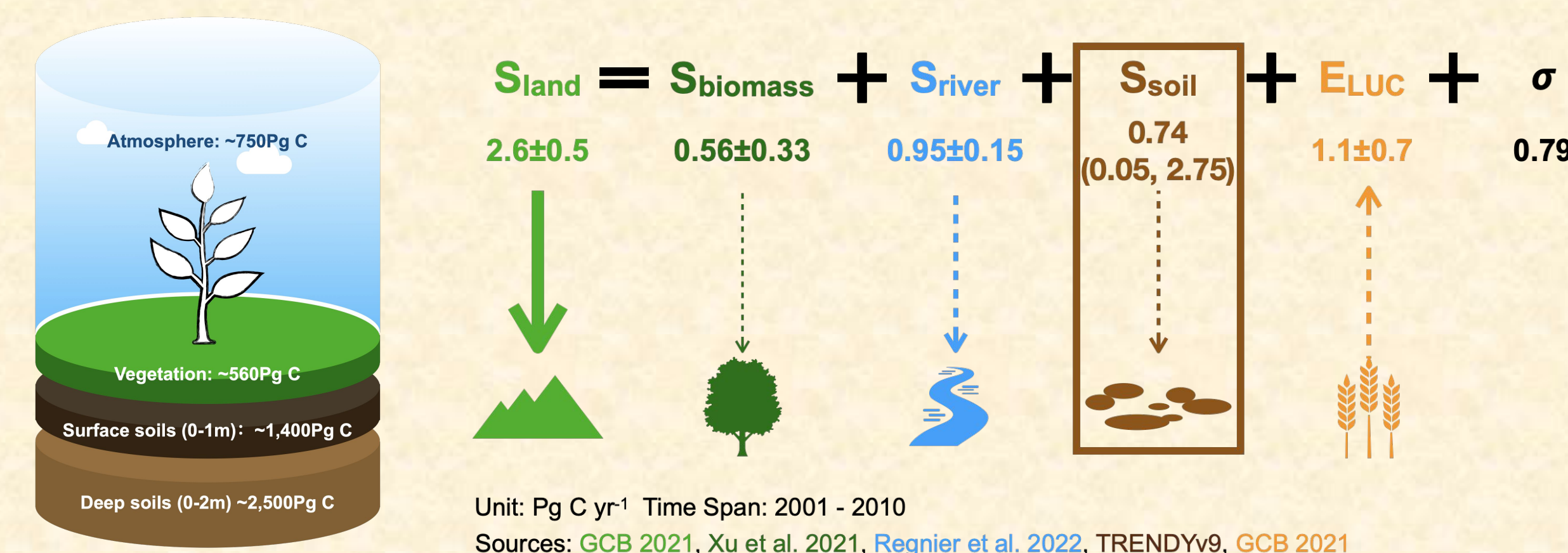
Computational Efficiency

- BINN is 30x faster than previous data assimilation approaches
- Matrix representation of CLM5
 - Allows for gradient-based optimization
 - Uses fast vectorized operations
- Distributed Data Parallel (DDP)
 - Save real physical time

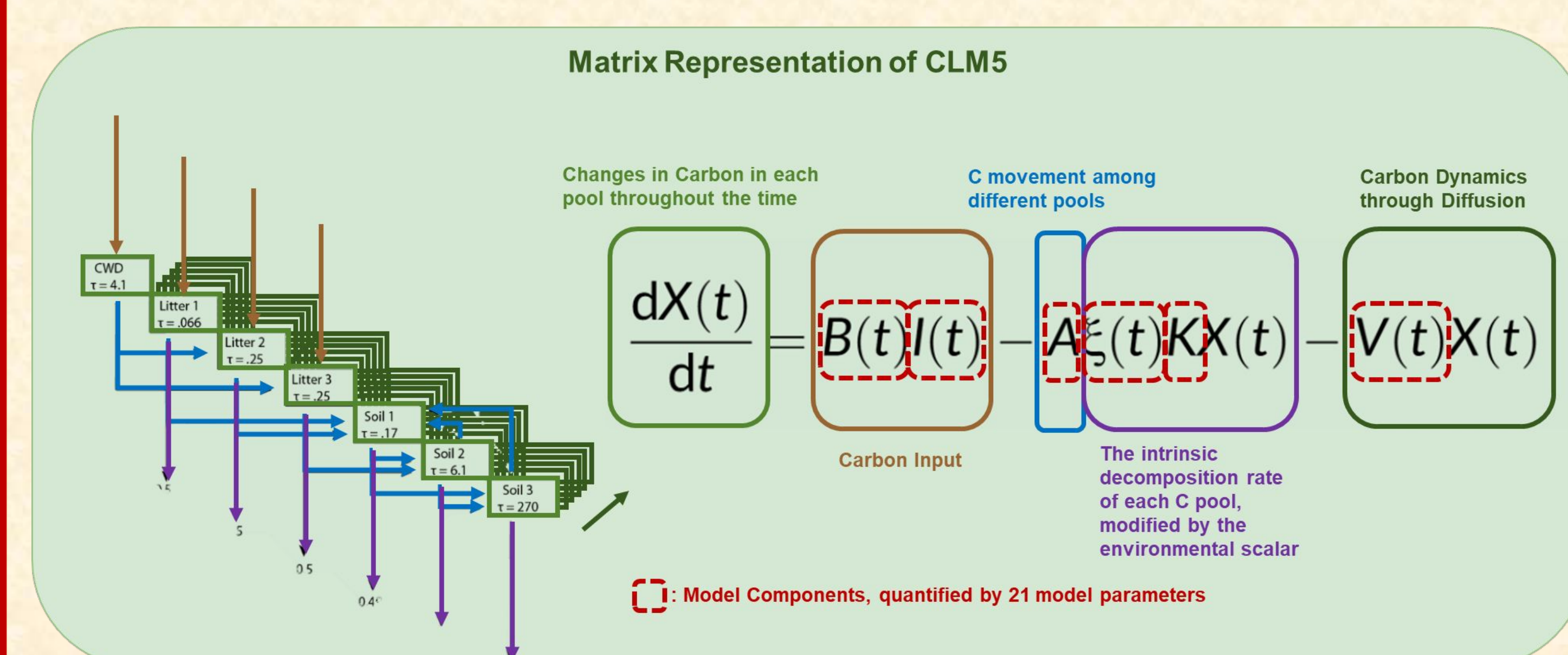


Modeling Soil Carbon Dynamics with CLM5

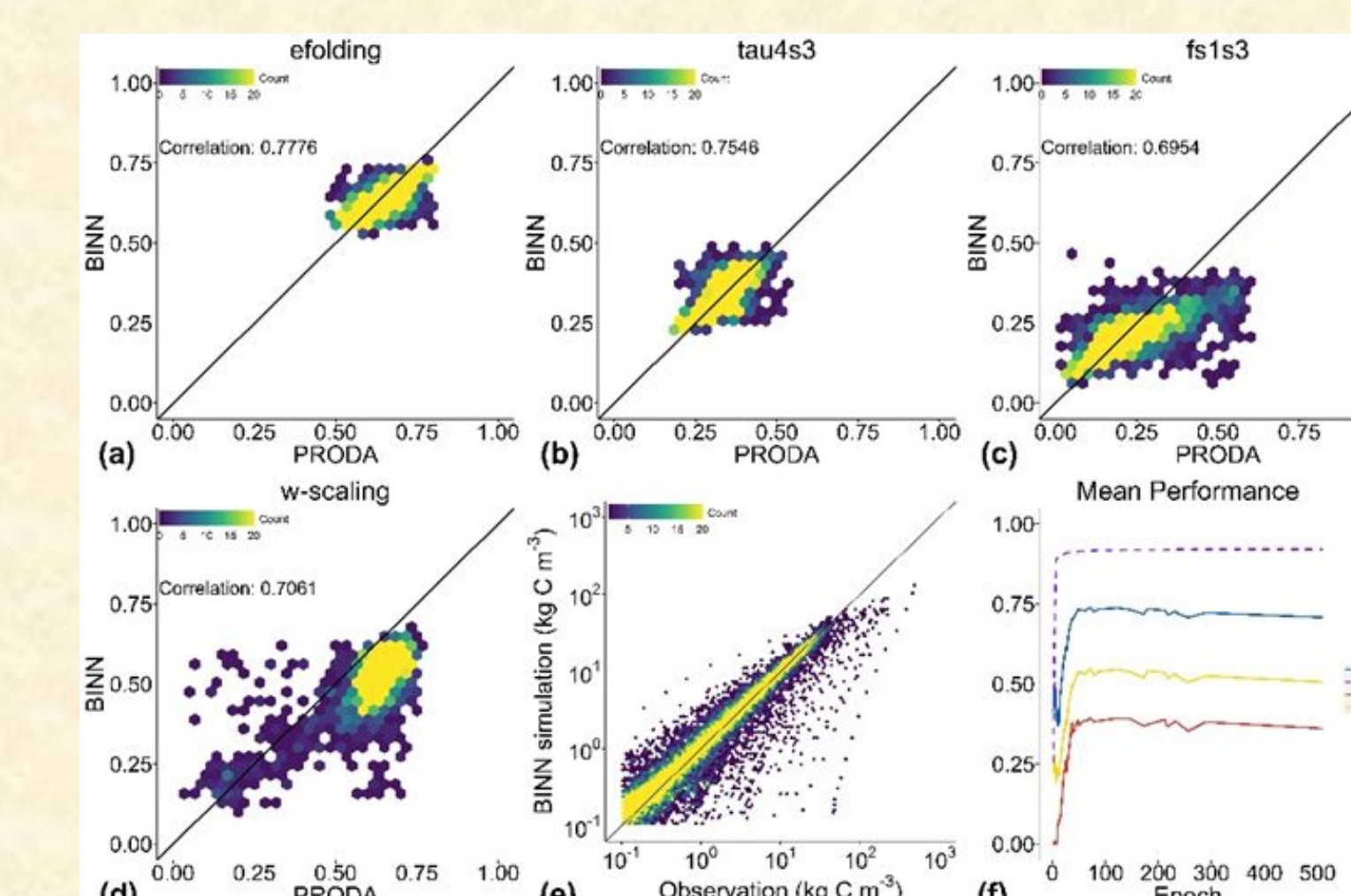
"Soils store more carbon than other terrestrial ecosystems."
(Tao et al. 2023)



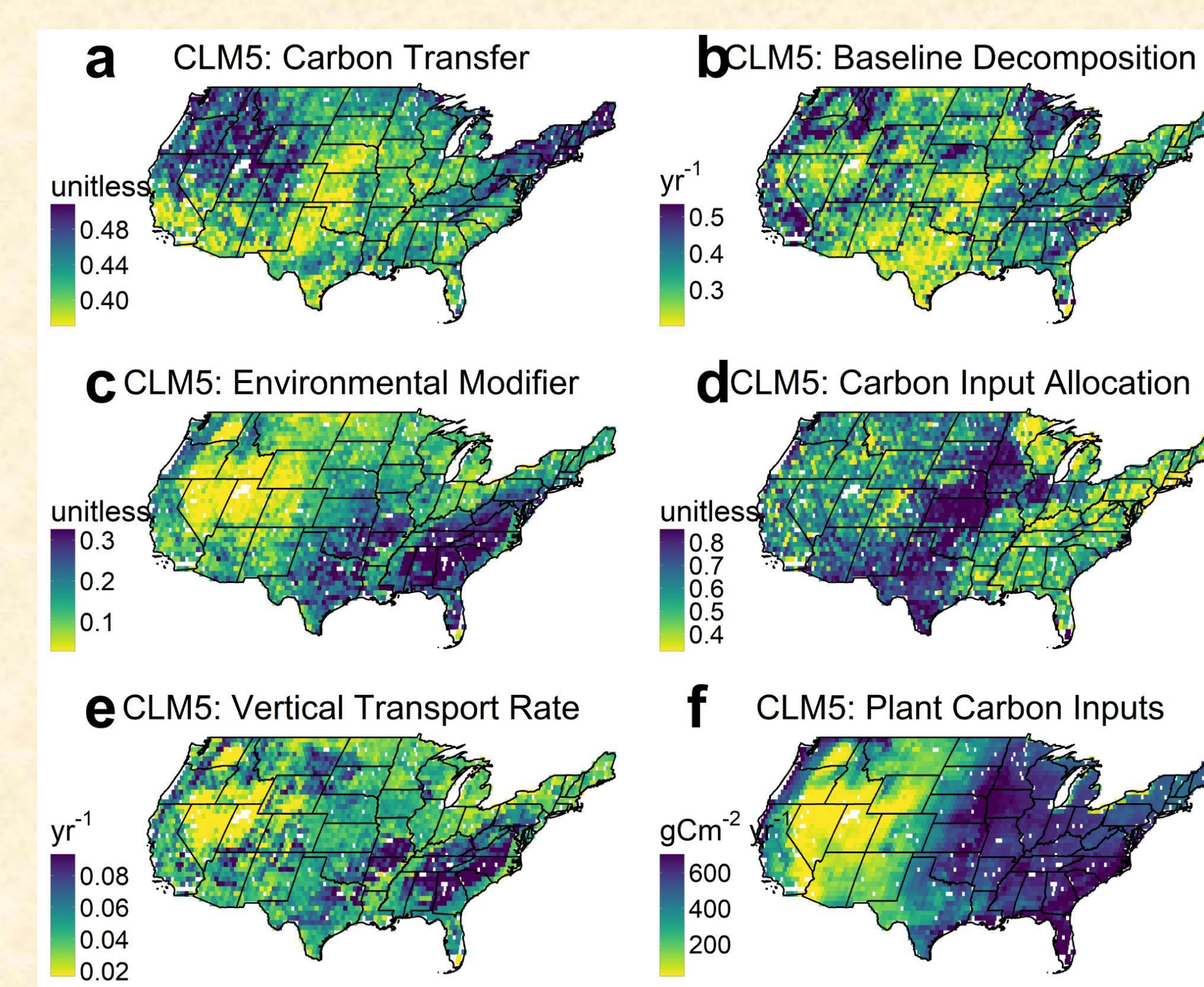
- Ecologists have developed **process-based models** to describe how carbon flows through different parts of the soil
 - Some processes are not fully understood → approximated by parameters that vary across space
- Example: Community Land Model 5 for soil organic carbon



Retrieve Emerging Mechanisms



BINN can retrieve the prescribed parameter values in a retrieval test (above) and discover emergent mechanisms from environmental information and SOC observations (below)



Toward more realistic projections of SOC

BINN simulated SOC shows less spatial bias across Contiguous US (left) and considerable accuracy (right)



Next Steps

- Try different model structures for enhancing scientific discovery and improving prediction accuracy
 - e.g. Graph Attention Networks
- Incorporate data from multiple sources
- Estimate parameter uncertainty
- Investigate how well BINN performs in low-data settings

References

Tao, F., Huang, Y., Hungate, B. A., Manzoni, S., Frey, S. D., Schmidt, M. W., ... & Luo, Y. (2023). Microbial carbon use efficiency promotes global soil carbon storage. *Nature*, 618(7967), 981-985.

Tao, F., Zhou, Z., Huang, Y., Li, Q., Lu, X., Ma, S., ... & Luo, Y. (2020). Deep learning optimizes data-driven representation of soil organic carbon in Earth system model over the conterminous United States. *Frontiers in Big Data*, 3, 17.